Functional Requirement Document

Here is the detailed Functional Requirement Document (FRD) based on the provided Business Requirement Document (BRD):

FR-SQG-001: Generate SQL Query for Sales Orders Data

• Title: Generate SQL Query for Sales Orders Data

• Description: The system must generate a complete SQL SELECT statement that applies transformation logic, includes appropriate joins, and ensures correct data types and formatting based on the provided tabular mapping document.

• Preconditions:

• The input mapping table is available with the required columns (Column Name, Column Type, Source Table, Source Column/Transformation Logic, Join Condition, Common Filter).

• The source tables (b_source.ord_hdr, b_source.ord_dtl, s_master.sale_orders, etc.) exist and contain the required data.

• Main Flow / Functional Steps:

1. **Create Intermediate Table (sales_orders_comp_stg)**:

• Retrieve the mapping document for sales_orders_comp_stg.

• Generate a SQL SELECT statement that applies transformation logic, joins, and filters as specified in the mapping document.

• Use aliases (a, b, etc.) consistently as per the join logic.

• Ensure correct data types and formatting (e.g., date conversions).

2. **Create Main Table (sales_orders_comp)**:

• Retrieve the mapping document for sales_orders_comp.

• Generate a SQL SELECT statement that applies transformation logic, joins, and filters as specified in the mapping document.

• Use aliases (a, b, etc.) consistently as per the join logic.

• Ensure correct data types and formatting (e.g., date conversions).

3. **Handle Conditional Logic**:

• Apply IF, CASE, and other conditional logic as described in the mapping document.

4. **Format the SQL Query**:

• Format the generated SQL query for readability (indentation, line breaks).

- Detailed Requirements:

- The system must generate a SQL query that includes all necessary joins and filters.
- The system must apply transformation logic as specified in the mapping document.
- The system must use aliases consistently as per the join logic.
- The system must ensure correct data types and formatting (e.g., date conversions).
- The system must handle conditional logic (e.g., IF, CASE) as described in the mapping document.

Some of the specific SQL requirements include:

- Joining tables based on specified join conditions (e.g., trim(a.referenceto)=trim(b.referenceto) and trim(a.refno)=trim(b.refno)).
- Applying transformation logic (e.g., CONCAT(referenceto,refno), cast(to_date(from_unixtime(unix_timestamp(cast(a.dateupd as string), 'yyyyMMdd'))) as date)).
- Using subqueries and derived tables as necessary (e.g., (select from b_source.ord_hdr where delete_flag <> 'D') a).
- Handling null values and default values as specified in the mapping document (e.g., SET TO NULL, SET TO BLANK).